

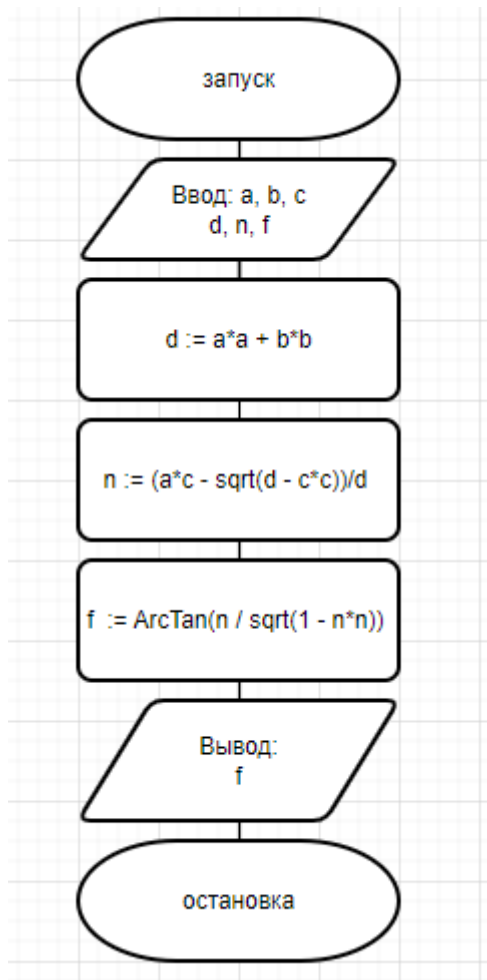
Задания для обязательного выполнения
по материалам Лекции № 2, Тема № 2

Задание 1.

$$a = \arcsin \left(\frac{AC - \sqrt{A^2 + B^2 - C^2}}{A^2 + B^2} \right)$$

1.

Алгоритм:



Программа:

```
var
  a,b,c : real;
  d, n, f : real;
begin
  readln(a);
  readln(b);
  readln(c);
  d := a*a + b*b;
  n := (a*c - sqrt(d - c*c))/d;
  f := ArcTan(n / sqrt(1 - n*n));
  writeln(f);
end.
```

```
•Program1.pas*

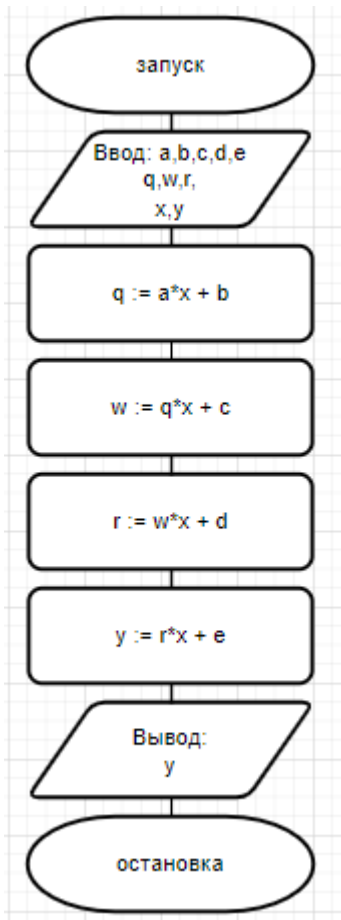
var
  a,b,c : real;
  d, n, f : real;
begin
  readln(a);
  readln(b);
  readln(c);
  d := a*a + b*b;
  n := (a*c - sqrt(d - c*c))/d;
  f := ArcTan(n / sqrt(1 - n*n));
  writeln(f);
end.
|

Окно вывода

10
7
2
0.0534375457435123
```

2. $y = ax^4 + bx^3 + cx^2 + dx + e = \{[(ax + b)x + c]x + d\}x + e.$

Алгоритм:



Программа:

```

var
  a,b,c,d,e : real;
  q,w,r : real;
  x, y : real;
begin
  readln(x);
  readln(a);
  readln(b);
  readln(c);
  readln(d);
  readln(e);
  q := a*x + b;
  w := q*x + c;
  r := w*x + d;
  y := r*x + e;
  writeln(y);
end.
  
```

```

•Program1.pas*
var
  a,b,c,d,e : real;
  q,w,r : real;
  x, y : real;
begin
  readln(x);
  readln(a);
  readln(b);
  readln(c);
  readln(d);
  readln(e);
  q := a*x + b;
  w := q*x + c;
  r := w*x + d;
  y := r*x + e;
  writeln(y);
end.
  
```

Окно вывода

2
2
3
4
5
6
88

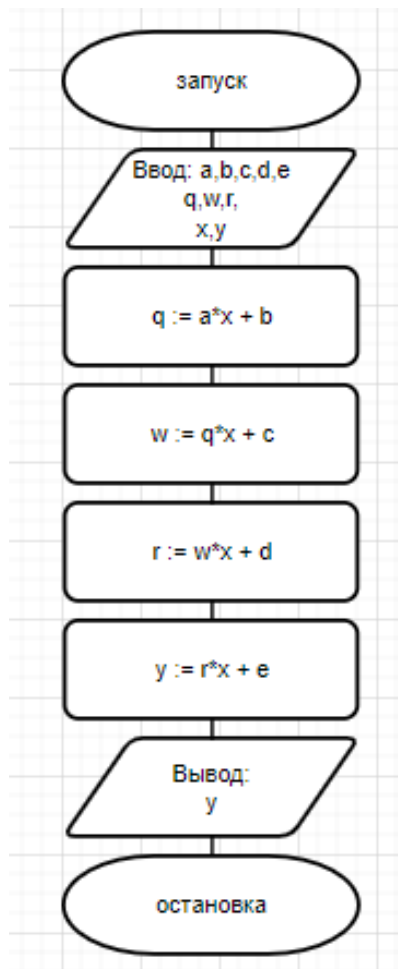
$$r = (f_2/f_1 - 1 + 0,707 f_2/f_1 \sqrt{1 - f_1/f_2})^2 =$$

$$= (f_2/f_1 (1 + 0,707 \sqrt{1 - f_1/f_2}) - 1)^2.$$

3.

Алгоритм:

Программа:



```

var
  f, t : real;
  r : real;
  q, w, e : real;
begin
  readln(f);
  readln(t);
  q := sqrt(1 - f/t);
  w := 1 + 0.707*q;
  e := t/f * w - 1;
  r := e*e;
  writeln(r);
end.
  
```

```

•Program1.pas*
var
  f, t : real;
  r : real;
  q, w, e : real;
begin
  readln(f);
  readln(t);
  q := sqrt(1 - f/t);
  w := 1 + 0.707*q;
  e := t/f * w - 1;
  r := e*e;
  writeln(r);
end.
  
```

Окно вывода

```

3
4
0.6474884444444444
  
```